

```
SELECT id, sales, RANK() OVER (ORDER BY sales DESC) AS rank
FROM my_table;
```

These features let you carry out complex data manipulations right in DuckDB, using SQL's powerful syntax. DuckDB is a flexible tool for loading, querying, and changing data from different formats. It works well with pandas to give a data analysis experience.

How to use DuckDB with big data

DuckDB is very efficient for working with large datasets because it can read data from the disk and stream it, which means it can query data that is too big to fit in memory. This is different from traditional in-memory databases, which read all the data from the disk.

Data that is bigger than memory: DuckDB can work with datasets that are bigger than your system's RAM by breaking them up into smaller pieces. When you run a SQL query on a large dataset stored in a file (like Parquet), DuckDB only reads the parts of the file that are needed for the query. This means that you don't have to load the whole dataset into memory, which makes it possible to do analytics on terabyte-sized data with modest hardware.

Streaming and out-of-core features: DuckDB is built to handle streaming data. It reads and processes data in a streaming way, so you can do complex things like joins, aggregations, or filtering on large datasets without using up too much memory. This makes it perfect for workflows that involve continuous data ingestion or real-time analysis, since DuckDB processes data in a streaming way.

Scanning Parquet files quickly: DuckDB is very good at quickly reading Parquet files. Because Parquet stores data in columns, DuckDB can only scan the columns it needs for a query, which cuts down a lot on I/O. For example:

```
SELECT column1, column2 FROM large_parquet_file WHERE filter_
condition;
```

This selective reading cuts down on disk access and speeds up query execution. DuckDB also supports predicate pushdown, which filters data even more at the file level, cutting down on unnecessary data reads.

Using DuckDB to speed up pandas workflows

DuckDB can speed up pandas workflows a lot, especially when working with big datasets. It does this by acting as a fast query engine for pandas DataFrames, letting users use SQL's expressive power to replace slow pandas operations.

Instead of using Python loops or complicated functions to go through DataFrames one row at a time, you can register pandas DataFrames in DuckDB and run SQL queries directly. This method takes advantage of DuckDB's optimised query execution,

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